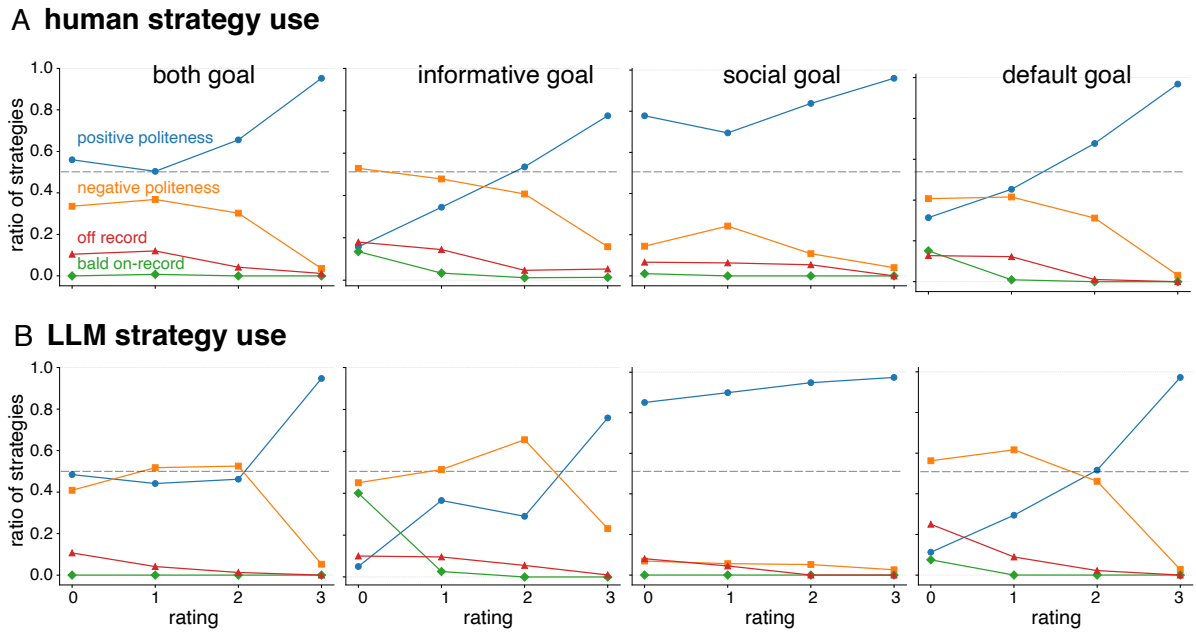

Figure 3: Proportion of different politeness strategies across ratings and goals for (A) human and (B) LLMs.

is employed in open-ended responses. Among all 1,248 responses collected, 527 (42.2%) used the specific pattern of adjective evaluation studied by Yoon et al. (2020). Within this subset, 35 responses (6.6%) employed negation as a politeness strategy, and negation was most common in low-rating (0 or 1 heart) scenarios, which qualitatively replicates our findings from Experiment 1 (see Figure 5 in Appendix for details). Thus, the negation strategies studied in constrained settings do appear in open-ended production, but represent just one of many politeness devices available to speakers.

5.2 Word usage patterns

To better understand differences between human and LLM responses, we analyzed unigram distributions using Pointwise Mutual Information (PMI) and Jensen-Shannon Divergence (JSD). First, examining unigrams with the highest PMI, we found that human responses more frequently incorporated casual language and expressions (e.g., “awesome," “great"), whereas LLM-generated responses tended toward more formal linguistic choices (e.g., “fabulous," “excellent"). Both groups effectively employed personalization as a politeness strategy, such as directly mentioning the performer’s name (e.g., “Your app is pretty good, Henry!"). Additionally, both humans and LLMs adapted their lexical choices based on context, with minimal overlap in high-PMI words across different goals and ratings.

Next, we quantified differences in empirical word frequency distributions by calculating the Jensen-Shannon Divergence (JSD). Interestingly, the JSD between the lexical distributions of preferred and non-preferred response groups was quite small (JSD = 0.013) though still significantly different than a permuted null distribution ($p < 0.001$), while all other group comparisons showed much larger differences (JSD > 0.13, $p < 0.001$; see Table 10). This suggests that simple lexical choice may not be the primary driver of human preferences in polite language.

Finally, we conducted higher-dimensional analyses using SBERT embeddings (Reimers and Gurevych, 2019) to distinguish between response categories. These analyses (described in Appendix C.2) revealed that while human vs. LLM responses were readily distinguishable in embedding space (83% accuracy), preferred vs. non-preferred responses were much harder to classify (54% accuracy). LLM responses were more distinguishable across different communicative goals than human responses, suggesting more stereotyped strategies.

5.3 Annotated politeness strategies

To obtain a comprehensive picture of the politeness strategies employed in human and LLM responses, we conducted a detailed annotation using the politeness framework from Brown and Levinson (1987), supplemented by markers from Danescu-Niculescu-Mizil et al. (2013). This framework distinguishes four broad categories of polite-

Category	Politeness Strategy	Examples
Positive Politeness	**Gratitude** **Be optimistic**	I'm so *grateful* that . . . You can even make them better next time!
Negative Politeness	**Apologizing** **Question or hedge**	I didn't like it, sorry! Maybe you can try something different?
Off-Record	**Be vague** **Give association clues**	It was *interesting*. (true rating is 0) It was better than those who can't play.
Bald on-Record	**Negative lexicon** **Factuality**	It was *terrible*! I didn't like the cookies at all.

Table 2: Examples of utterances for different politeness strategies (two per category), with the corresponding sub-strategy highlighted in bold. See the comprehensive list in Table 12.

ness strategies with many subtypes (see Table 2 and Appendix Table 12): positive politeness (e.g. compliments and expressions of interest), negative politeness (e.g. hedging and indirectness), off-record strategies (indirect hints that maintain plausible deniability), and bald-on-record strategies (direct statements without politeness). We used LLMs (`GPT-4.1` and `Claude-3.7-Sonnet`) as annotators, following best practices (Tan et al., 2024). Annotations were manually verified and corrected where necessary. We used LLMs (`GPT-4.1` and `Claude-3.7-Sonnet`) as annotators, following best practices (Tan et al., 2024). Annotations were manually verified and corrected where necessary. Appendix C.3 provides full details on the annotation process and framework.

Overall strategy distribution. As shown in Figure 3, both humans and LLMs rely primarily on positive politeness (rapport-building) and negative politeness (minimizing imposition) strategies, with relatively low use of off-record and bald-on-record approaches. However, a key difference emerged in strategy selection patterns: while both humans and LLMs increased their use of positive politeness strategies as ratings increased, LLMs showed systematically higher use of negative politeness strategies even in positive contexts (higher ratings), where humans tended to reduce such strategies. These strategy distributions were significantly different under a permutation test (JSD = 0.023, $p < 0.001$). This pattern, where LLMs maintain high levels of hedging and indirectness across contexts, may reflect training objectives that prioritize avoiding potential harm over more human-like affirmation strategies.

Goal and rating sensitivity. We observed that both humans and LLMs appropriately varied their strategy distribution by communicative goal, with the informative goal showing the most distinct pattern. For the informative goal with high ratings (2-3 hearts), LLMs showed unexpectedly higher use of negative politeness strategies compared to humans, who shifted toward positive strategies in these contexts. This pattern suggests that LLMs may overuse hedging, conventional indirectness, and other distancing strategies even when giving positive feedback, potentially explaining some of the stylistic differences observed in the evaluation. However, as with word usage patterns, the differences between strategy distributions for preferred and non-preferred responses were not significant (JSD = 0.008, $p = 0.087$), suggesting that preference judgments may be driven by higher-order social factors beyond the mere presence or absence of specific words or politeness strategies.

Cross-cultural variation. Given documented cultural differences in politeness norms, we also conducted an exploratory analysis testing whether strategy distributions varied between our US and UK participants. We found a statistically significant difference (JSD = 0.5279, $p < 0.001$). US participants deployed a more diverse range of sub-strategies overall than UK participants, especially among positive strategies; however, the effect size was relatively small and our sample was imbalanced (532 US responses vs. 92 UK responses). While this preliminary finding suggests some cross-cultural variation even within English-speaking populations, more balanced sampling would be needed to characterize these differences systematically. Future work should examine how cultural

context shapes both human and LLM politeness strategies across a broader range of English varieties and other languages.

6 Discussion

We find that LLMs demonstrate impressive pragmatic competence in capturing politeness phenomena, but also differ in important ways from humans. Most notably, in open-ended production, LLMs continue to rely on negative politeness strategies (hedging, indirectness) even in positive contexts where humans shift toward positive strategies (rapport-building). Interestingly, LLM responses were consistently preferred by human evaluators over crowd-sourced human responses, suggesting that preference judgments may track different features of the utterance than the mere ability to reproduce human-like patterns. This contrast — aligning with human patterns in constrained multiple-choice tasks while diverging in more open-ended tasks — emphasizes that LLMs leverage a richer repertoire of strategies when given the opportunity to express them, such as the familiar "COMPLIMENT, but CRITICISM" constructions (Prochazka et al., 2020) we observed in low-rating scenarios, which were not provided as available options in the constrained tasks.

Recent developments in politeness theory may offer insight into these results. While Brown and Levinson (1987) classically emphasized the role of face threat, subsequent work has reconceptualized politeness as part of a broader class of *relational work* (Spencer-Oatey, 2011; Locher, 2013). From this perspective, politeness strategies manage the affective quality of relationships through multiple components: face concerns (desires for positive evaluation and acknowledgment of identity), sociality rights (what people expect from each other in interaction, including expectations about appropriate levels of imposition and warmth), and other relational goals.

Our findings suggest that humans and LLMs may functionally assign different weights to these components. Humans emphasize equity rights (avoiding imposition) when delivering criticism but shift toward association rights (building warmth) when giving praise. But LLMs consistently maintain a high weight on imposition across all contexts, potentially aligning with user expectations for AI systems, where maintaining appropriate distance could be more desirable than pursuing interpersonal warmth. Understanding how different training objectives and alignment procedures implicitly shift these relational weights represents an important direction for future work, particularly so developers can make intentional choices about more or less desirable interpersonal dynamics.

Distributional differences in politeness strategies may also have practical implications for human-AI communication. Following Gricean principles of communication as rational social action (Grice, 1975), listeners expect speakers to modulate their politeness strategies based on the context: you shouldn't need to take steps to mitigate face threat if face isn't threatened. So when AI systems violate these expectations, they create an interpretive puzzle: users must determine whether hedged language reflects the literal content being communicated or whether it is an artifact of the system's persona, risking pragmatic misinterpretations where humans interpret hedged positive feedback as more negative than intended. For example, does "your analysis seems reasonably sound" mean there are specific weaknesses that ought to be addressed, or is this just how the system expresses unqualified approval? Users may adapt their expectations of the system over time (Branigan et al., 2010; Araujo and Bol, 2024; Zhou et al., 2023; Waytz et al., 2014), but may still incur a hidden cognitive cost associated with correcting for these differences. Future work should examine how these differences play out across cultural contexts, where politeness norms already vary substantially (Wierzbicka, 2003; Ide, 1989).

In conclusion, our findings provide an empirical foundation for understanding differences between humans and AI systems in a key domain of pragmatic language use. Critically, our study was descriptive rather than prescriptive. Divergences matter not because artificial simulacra of human-like behavior is desirable, but because it affects human-AI communication in practice, raising questions about how users learn to interpret AI-generated politeness differently than human-generated politeness. Beyond practical applications, these findings also contribute to theoretical debates about the cognitive architecture underlying polite speech, and the computational principles that guide decisions across social contexts. These findings could inform future LLM training approaches to better align with human pragmatic patterns, laying groundwork for making intentional rather than accidental choices about subtle pragmatic patterns in AI systems.